

VU Medical Zone

BT 302 Current Final Term Updated Spring 2021

Q1. Functions of Cytokines? 5 Marks

- Used as biological response modifier for treatment of various diseases
- Cytokine is a general term but more specific term used according to their cell of origin
- Monokines: produced by mononuclear phagocytic cells
- Lymphokines: produced by activated lymphocytes

Cytokines function as cascade signaling in the form of cytokine network

Cytokines are considered as redundant

- Redundancy: is due to the nature of cytokine receptors

Q2. Subregion of variable region of antibody? 5 Marks

Variable region

- Present on heavy & light chains of Immunoglobulins

Two sub-regions

1) Hypervariable (HVR)

- Based on variability of amino acid sequences
- Complementarity determining region

2) Framework Regions

- Regions between hypervariable regions
- Groups & sub-groups of immunoglobulins
- Products of different variable region genes

Q3. Types of Autoimmunity? 2 Marks

Autoimmunity:

Classification based on tissues or organs involved

- Two following categories

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- Organ-specific autoimmunity
- Non-organ Specific autoimmunity

Q4. Effector Function of complement protein? 3 Marks

Effector Functions of Complement Proteins

- Opsonization (Opsonins)
- Chemoattraction (Chemoattractants)
- Anaphylaxis (Anaphylatoxins)
- Pro-Inflammatory

Q5. Primary and secondary antibody response? 2 Marks

Primary Ab response

- First kind of antibody response against antigen
- Major class of antibody is IgM
- Ab are with less affinity & less persistent

Secondary Ab response

- Second kind of antibody response against antigen
- Major class is IgG but may IgE or IgA but less IgM

Q6. Inflammation and its role in body? 3 Marks

Inflammation is the body's attempt: To protect the body from harmful stimulus. Complex Tissue response

- Part of body's immune system
- Inflammatory response: series of events for inflammation
- For wound healing
- For clearance of infections

Q7. What is Opsonization? 2 Marks

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Oponization: some complement proteins are good opsonins. C3b, iC3b and C4b attach to microorganism first. This complex of complement protein binds with complement receptor on the surface of phagocytic cell. Phagocytosis of microbe.

Q8. Write two main types of tumor associated antigens.

2 Marks

- There are two main types of tumor associated antigens

- 1) Tumor specific transplantation antigens

- 2) Tumor Associated transplantation antigens

Q9. Differentiate between Organ Specific autoimmunity and Non-Organ Specific autoimmunity. 2 Marks

Organ Specific autoimmunity

- Immune response against organ associated specific antigens
- Antibodies against organ associated antigens damage the organ
- Target organs
- Skin
- Thyroid gland
- Muscles

Non-Organ Specific autoimmunity

- Immune response not against organ associated specific antigens
- Antibodies against not organ associated antigens
- Systemic autoimmune diseases
- Mainly target organs: skin, joints, soft tissues etc.

Q10. Write down the mechanism of Oxygen independent killing? 3 Marks

- 1) Oxygen Independent

- No need of oxygen for such kind of intracellular killing of bacteria
- Granules & Vesicles of phagocytes secrete hydrolytic proteins

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- Those proteins are bacteriocidal in nature according to their modes of action

Effector Molecule Function:

Cationic proteins (including cathepsin): Damage to microbial membranes

Lysozyme: Splits mucopeptide in bacterial cell wall

Lactoferrin: Deprives proliferating bacteria of iron

Proteolytic and hydrolytic enzymes: Digestion of killed organisms

Q11. How the fragment of antibodies is obtained? Briefly elucidate the properties of the fragment? 5 Marks

Antibodies Fragments

- Produced by proteolytic enzymes i.e papain
 - Two fragments are produced which have structure & function relationship
- 1) Fab
 - Fragment for antigen binding (Fab)
 - Papain causes the cleavage at hinge regions to produce two identical fragments containing one heavy & one light chain
 - Having monovalent valency for antigen binding
 - 2) Fc
 - Crystalizable Fragment (Fc)
 - Contain two heavy chains with CH2 & CH3 domains
 - Performed effector functions

Q12. Applications of Agglutination reaction? 2 Marks

Applications of Agglutination

- Blood grouping: ABO & Rh for both either on the basis of antigens of RBC or antibodies of serum
- Assessing bacterial infections e.g Widal test for diagnosis of Typhoid fever

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Q13. Tolerance is achieved by natural and artificial? 2 Marks

Naturally: In case of autoimmune disorders e.g Rheumatoid Arthritis, SLE etc.

2) Artificially: exposure to immunosuppressive drugs or X-ray irradiation e.g in case of experimental animals for bone marrow transplantation

Q14. Usage of Monoclonal antibodies? 3 Marks

- Usage in prevention, diagnosis & treatment of disease
- Immunophenotyping of Immune cells: Monoclonal Ab used against cell surface molecules like Cluster of Differentiation (CD)
- Typing of various tumors e.g Leukemia • Immunohistochemistry of solid tumors
- Use in Immunotherapy: Against various tumor marker e.g CD20 B-Cell lymphoma
- Use in Fluorescence Activated cell sorting (FACS)

Q15. Cytokines in adaptive immune system? 5 Marks

- Cytokines which act primarily for Adaptive (Specific) immune system
- 1) IL-2
- Produced by Th cells but Tc also produced in lesser extent
 - Stimulate B-cell division Can also activate NK cells s& monocytes
 - IL-2 acts in autocrine manner on T-cells
 - Activation of T-cells result in expression of IL2R & production of IL-2
 - IL-2 binds to IL-2R and promotes cell division
 - IL-2R decay if T-cells are not activated & eventually proliferative phase ends
- 2) IL-4
- IL-4 is produced by macrophages & Th2 cells
 - Promotes the development of Th2 from naïve Th cells
 - Differentiated Th2 cells result in the production of antibodies
 - IL-4 involves in the class switching of antibody into IgE.
- 3) IL-5

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- IL-4 is produced by Th2 cells
 - Promotes the development of B-cells & eosinophiles
 - Also activates mature eosinophiles
- 4) TGF- β
- TGF- β is produced by T cells and many other cells types
 - Acts as inhibitory cytokine
 - Inhibits the proliferation of T-cells & activation of macrophages
 - Also acts on neutrophils & endothelial cells to block the effects of proinflammatory cytokines

Q16. Clonal Selection of Lymphocyte. 5 Marks

- T-Cell Receptors (TCR)
 - Receptor Specificity
 - Lymphocyte Repertoire
 - After maturation, selection of lymphocytes which are non-reactive to self-antigens
 - Four principles
- 1) Unique Receptor Specificity
 - 2) Lymphocyte Activation: Interaction with foreign molecule
 - 3) Differentiation of lymphocyte: from single clone
 - 4) Selection of lymphocytes bearing receptors against non-self-molecules

Q17. Immunization. 2 Marks

Ways of providing specific protection against many common & damaging pathogens.
Stimulation of organism's or individual's immune system.

Q18. Modes of Immunization. 2 Marks

These modes of immunization can be natural or artificial in their nature

- 1) Active Immunization
- 1) Passive Immunization

Q19. T-cell Receptor and B-Cell Receptor? 3 Marks

T-cell Receptor:

Surface receptor for binding with Ag bounded with MHC. Like immunoglobulin: Part of Ig Superfamily. TCR is a heterodimer surface receptor. Composed of two chains of equal length

1. A Chain
2. B Chain

Short Cytoplasmic tail: Not sufficient for signal transduction. Transmembrane region. for anchoring cell membrane (due to hydrophobic amino acids).

B-Cell Receptor:

Cross linking of BCR by Ag. By cross linking, there is activation of Src for cellular signaling. Src phosphorylates tyrosine on ITAM of $I\alpha$ & β . Phosphorylated ITAM act as docking site for Syk. Downstream signaling for antibody synthesis.

Q20. T-cell antigen Kinetic Process? 3 Marks

T-independent Ag

- Mainly IgM production
- All those phases in primary response
- No secondary response as same phases in second time exposure to Ag
- No memory response

Q21. Drug induced hemolytic anemia? 5 Marks

- Also known as cytotoxic hypersensitivity
- Effect various organs & tissues (Tissue specific)
- Antigens are usually endogenous
- Exogenous: Chemicals in the form of haptens
- Binds to the surface of cell membrane
- Antibodies against antigens: IgM or IgG

Example of exogenous, antigen: Drug induced hemolytic anemia, granulocytopenia & thrombocytopenia

- Ag-Ab reaction on the surface of RBC

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- Complement fixation on immune complex on RBC surface
- Reaction time is minutes to hour
- Lysis of RBC

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